Objective and constraint geometry

Composite SFRAMES template

$$\min_{x \in C} f(x) + g(Tx), \qquad x = (U, V), \qquad U, V \in \mathbb{R}^{512 \times 64}.$$

The reconstructed attenuation image is X = UVᵀ on a 512 × 512 grid. Every measurement view of one slice constrains the same two factors. Increasing the number of views changes the loss terms and leaves the variables fixed.

$$C = \{(U, V) : \ \|U\|_F \le R_U, \ \|V\|_F \le R_V\},$$

$$D = \{(U, V) : \ U \ge 0, \ V \ge 0\}, \quad g = \iota_D, \quad T = \mathrm{Id}.$$

The inequalities in D are entrywise. Its indicator g is zero on D and infinite elsewhere. C uses separate Frobenius balls. T acts on the factors and is the identity. The fan-beam projector acts inside f on their product.

Finite-sum smooth term f

$$q_i(U, V) = \frac{\|P_i(UV^\top) - y_i\|_2^2}{2d}, \qquad d = 956,$$

$$f(U, V) = \frac{1}{N} \sum_{i \in \mathcal{I}_{\text{train}}} q_i(U, V), \qquad N = |\mathcal{I}_{\text{train}}|.$$

Pᵢ is the calibrated fan-beam forward operator for a single angle. yᵢ contains the corresponding binned, log-corrected low-dose measurements. Each view and each detector bin have equal weight. Bilinearity makes the factor-space objective nonconvex.

Sampling and gradient

Each SFRAMES update draws a fresh, uniform subset B of 64 training views without replacement within the batch and independently across updates. All 956 detector bins in each selected view contribute. The batch gradient at a fixed iterate is unbiased for ∇f. Momentum averages these batch gradients.

$$E_B = \frac{\sum_{i \in B} P_i^* (P_i(UV^\top) - y_i)}{64d}, \qquad \widehat{\nabla f}_B = (E_B V, \ E_B^\top U).$$

Each joint gradient computes one subset forward projection and one subset adjoint. The polynomial loss is smooth on compact C. Global smoothness on the ambient space would require an extension outside a neighborhood of C.

Measured CT data and view splits

2DeteCT acquisitions

We fit real low-dose Mode 1 measurements from the 2DeteCT fan-beam collection. A separately acquired high-dose Mode 2 reconstruction supplies an image-quality reference for each slice.

Phase	Slice IDs	Training views	Role
Development	04300	360	Select one rate per method
Evaluation	04531, 04800	360 and 1,080	Reuse selected rates

Preprocessing and geometry

Starting from the 3,601 × 1,912 Mode 1 sinogram, discard the repeated final angle and sum adjacent detector cells. Apply the same binning to the dark field and the mean of the two flat fields. Apply −log to the dark-subtracted transmission ratio after clipping the ratio below at 10^{-6}, then reverse the detector axis. The clipping count is zero on these slices.

ASTRA 2.5.0 uses its CPU line_fanflat projector with physical lengths in mm. Source-to-origin is 431.019989 mm, origin-to-detector 97.980499 mm, binned detector spacing 0.1496 mm, and field width 116.528143 mm. The field is centered at (−0.113797, −0.113797) mm. Angle i is $\pi - 2\pi i/3600$.

Align the Mode 2 reference by vertically flipping its published 1024 × 1024 TIFF, dividing by the native 0.113797 mm voxel width and averaging 2 × 2 blocks onto the fitting grid. Evaluate relative image error against this reconstruction.

Exact view indices

Indices run from 0 to 3599. The 360-view training set uses i mod 10 = 0. The 1,080-view training set uses residues 0, 3, and 6. All runs hold out residue 9 for validation, exactly 360 views. Calibration uses i mod 30 = 0 and therefore the same 120 training angles at both view counts.

Data provenance

The processed inputs and TIFF-level checksums are recorded in the experiment's data/slice*.json sidecars. Slices 04300 and 04800 came from the 2DeteCT Zenodo archives. Slice 04531 came from DeepInverse's single-slice mirror.

2DeteCT data paper

DeepInverse preprocessing and geometry example

Slice 04531 single-slice archive

SFRAMES schedules and paired starts

Open-loop updates

$$\gamma_k = \gamma_0(k+1)^{-3/4}, \quad \rho_k = (k+1)^{-1/2}, \quad \mu_k = \mu_0(k+1)^{-1/4}.$$

Iteration k starts at zero. Use the first sampled gradient to initialize momentum. Thereafter blend the current batch gradient into momentum with weight ρ_k. Add the Moreau penalty gradient after the momentum update and apply the Frank–Wolfe convex combination.

$$s_k = \mathrm{LMO}_C\left(m_k + \frac{x_k - \Pi_D(x_k)}{\mu_k}\right), \qquad x_{k+1} = (1-\gamma_k)x_k + \gamma_k s_k.$$

Π_D clips negative entries. For a block direction G and radius R, the Frobenius ball LMO returns $-RG/\|G\|_F$ for nonzero G and zero when G is zero.

Slice-level calibration and rates

Slice	R_u	R_v	L_u	L_v	μ_0	γ_0
04300	4.12869674	4.12867862	0.19331899	0.19443450	5.14312016	0.10 and 0.25
04531	3.81305622	3.81304219	0.16367101	0.16325009	6.10981743	0.25
04800	4.22469276	4.22465958	0.20220242	0.20283452	4.93012733	0.25

For the feasible start, divide the positive backprojection by the sensitivity P*P1 floored at 10^{-3} of its maximum. Factor this image with rank-64 NNDSVD. Add 10^{-3} times each factor's RMS times independent uniform [0,1) draws. Fit a common least-squares amplitude to the calibration views with floor 10^{-6}, then scale each factor by its square root. Each cap is twice the resulting factor norm.

Estimate the image-space curvature λ using 10 power steps on P*P for the 120 calibration views and divide by 120d. Use $L_u = \lambda\|V_{cal}\|_{op}^2$ and $L_v = \lambda\|U_{cal}\|_{op}^2$, then set $\mu_0 = 1/\max(L_u, L_v)$. These are empirical initial curvature scales.

Paired initialization and random streams

The rerun keeps the original caps and calibration scales. Draw two independent standard Gaussian block directions and apply one Frobenius LMO to each. This starts on the boundary of C and outside D. It also removes the original data-informed warm start. Both variants start with empty gradient momentum.

Base seed is 110 for development and 201 for evaluation. NumPy's independent minibatch stream uses base + 100,000 in both variants. The LMO direction stream uses base + 200,000. The two evaluation view counts share the same LMO factors within each slice.

Schedule source: Improved convergence and stochastic estimation for the Frank-Wolfe with Moreau Envelope Smoothing (FraMES) algorithm, local manuscript dated 27 September 2026, Corollary 2.10 and Proposition 2.11.

Work budgets and comparator settings

Run budgets and diagnostics

Phase	Target views	Probe spacing	SFRAMES endpoint
Development	5,760	720 views	90 updates / 5,760 views
Evaluation	21,600	2,160 views	338 updates / 21,632 views

Stop after the first completed update reaching the target. Save at iteration 0, after update 1, when crossing a probe threshold and at the endpoint. A forward view and a backprojected view are counted separately. Optimizer time includes projections and oracle calls and excludes calibration, diagnostics and checkpoint I/O.

Feasibility repair clips negative entries and rescales each block if needed. Repaired training loss, held-out loss and reference error use those factors. Raw loss uses the actual iterate. Cap excess and distance to D describe the raw factors. Diagnostics leave optimizer state unchanged.

Original four-method pilot

All four methods share the original feasible warm start. Each development grid had two rate multipliers. The winner had the smaller repaired training loss at the last probe within 5,760 forward views. The LMO ablation changes only SFRAMES' initialization and keeps the original rates.

Method	Development rates	Selected	Step rule
SFRAMES	0.10, 0.25	0.25	Rate is γ_0. $\rho_0 = 1$.
SPRING-SARAH	0.30, 0.70	0.70	Block steps rate/L_u, rate/L_v.
ProxSARAH	0.40, 0.90	0.90	Joint step rate/($L_u + L_v$).
PALM	0.60, 1.10	1.10	Block steps rate/L_u, rate/L_v.

SPRING-SARAH alternates blocks with independent full refresh probability 1/16 per block and uses 64-view SARAH differences. ProxSARAH uses a joint step, 256-view snapshots, 12 recursive updates per snapshot and relaxation 0.8. Its recursive batches also have 64 views. PALM uses the full training set for each block step. These are the practical settings tested on this pilot.

Elite ran ASTRA forward and adjoint operations in FP32 with FP64 factor arithmetic and one BLAS thread per worker. Timing reflects concurrent arms on separate physical cores of the same host.

SPRING, Algorithm 1.1

ProxSARAH, Algorithm 1 and §3.2

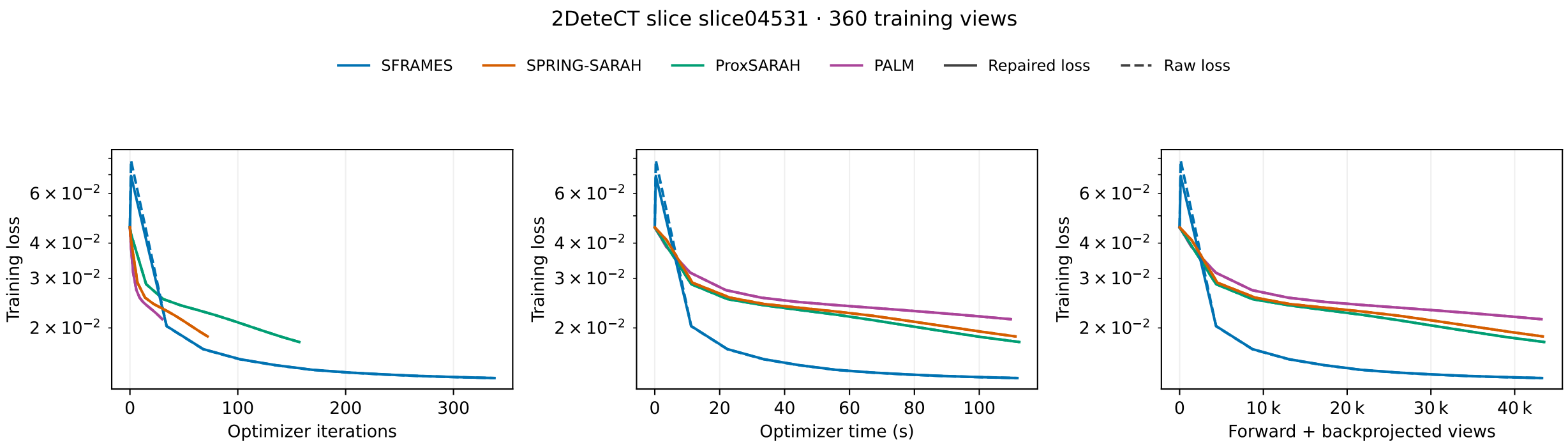
PALM, original paper

Original four-method evaluation

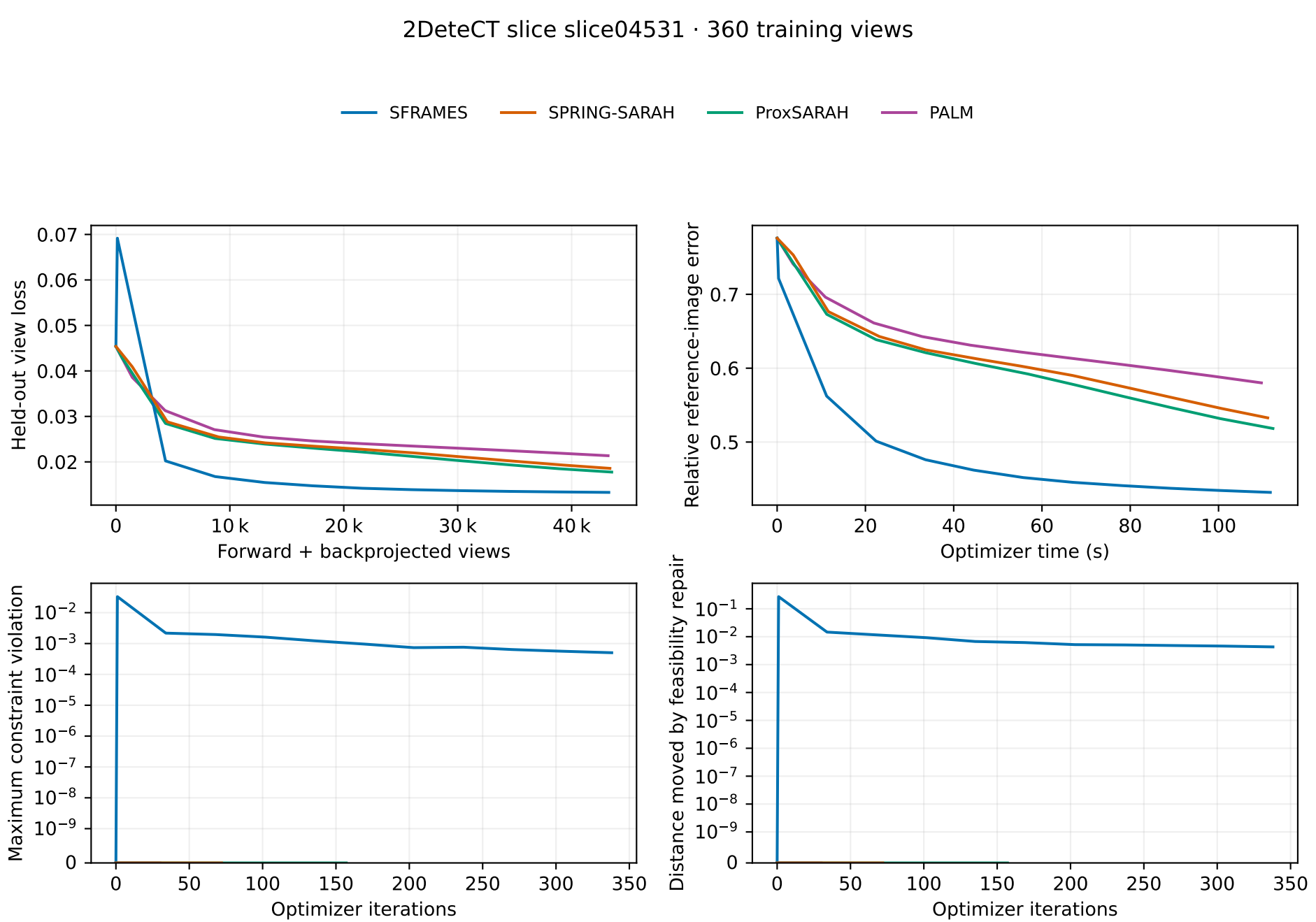
All methods began at the same D-feasible point for each evaluation case.

Method	Repaired training loss	Held-out view loss	Relative reference error
SFRAMES	0.01572	0.01577	0.45174
SPRING-SARAH	0.02448	0.02450	0.59124
ProxSARAH	0.02232	0.02235	0.55812
PALM	0.02730	0.02733	0.62592

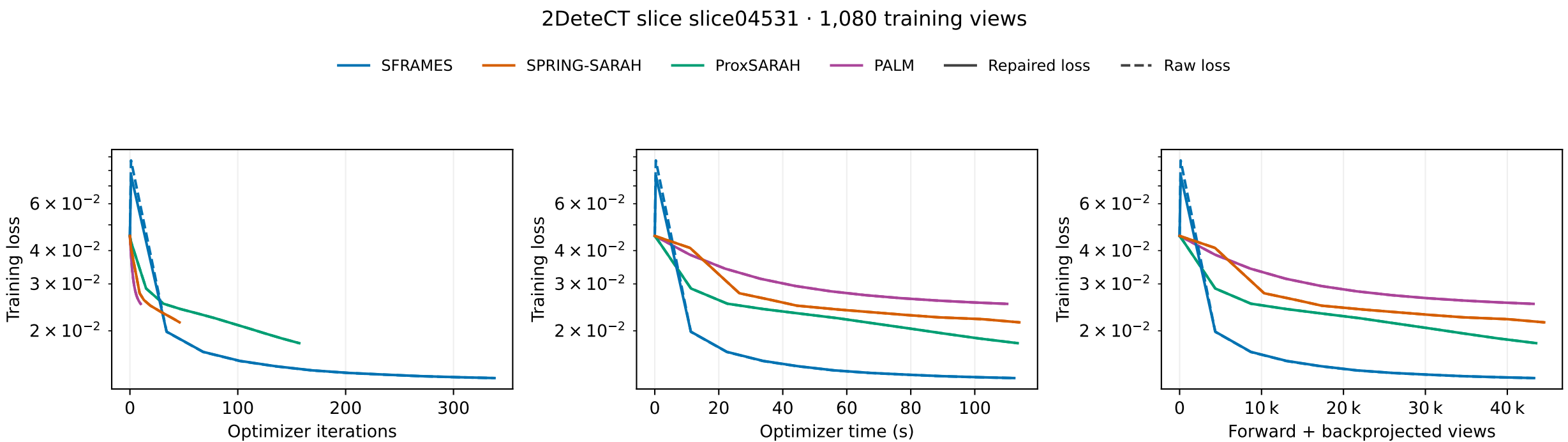
Medians across evaluation cases. Each value uses the last saved probe within its forward-view budget.
The next pages show the saved trajectories for each case.



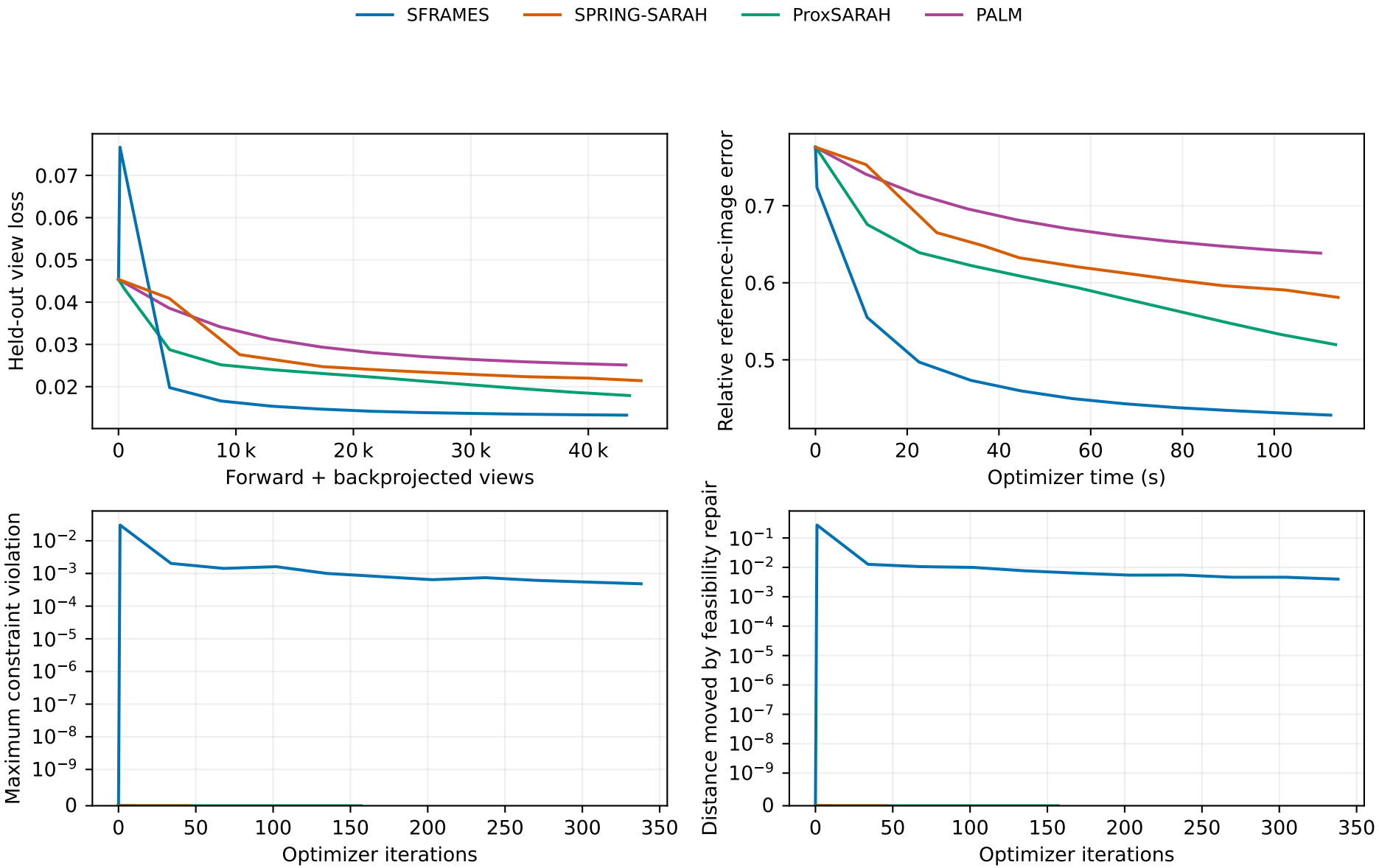
View counts include optimizer refreshes and recursive gradient differences. Timing excludes diagnostic probes and checkpoint I/O.



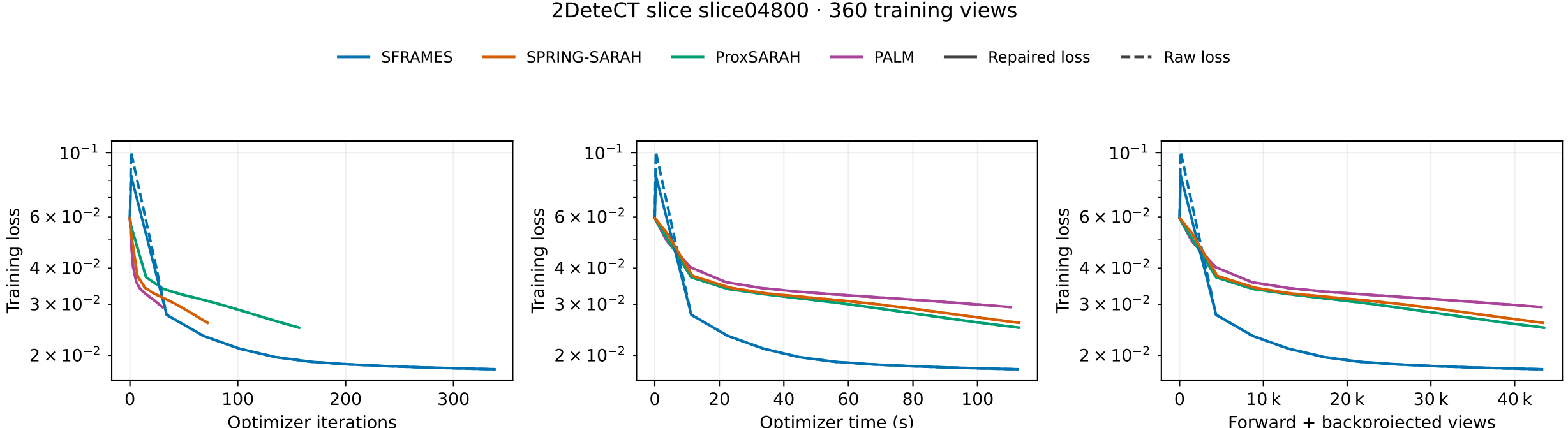
Held-out loss and reference error use repaired factors. Constraint violation and repair distance describe the raw iterate.



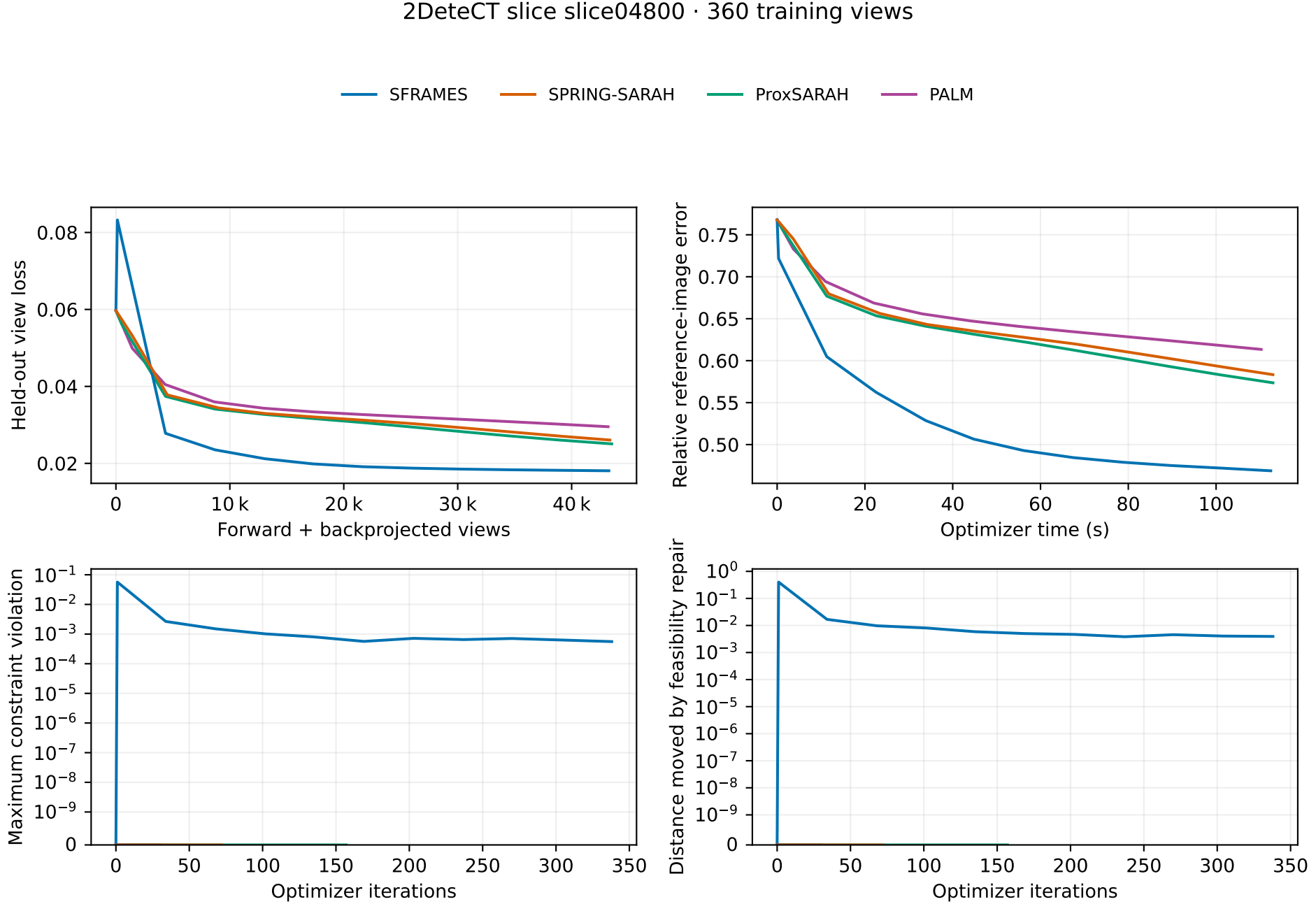
View counts include optimizer refreshes and recursive gradient differences. Timing excludes diagnostic probes and checkpoint I/O.



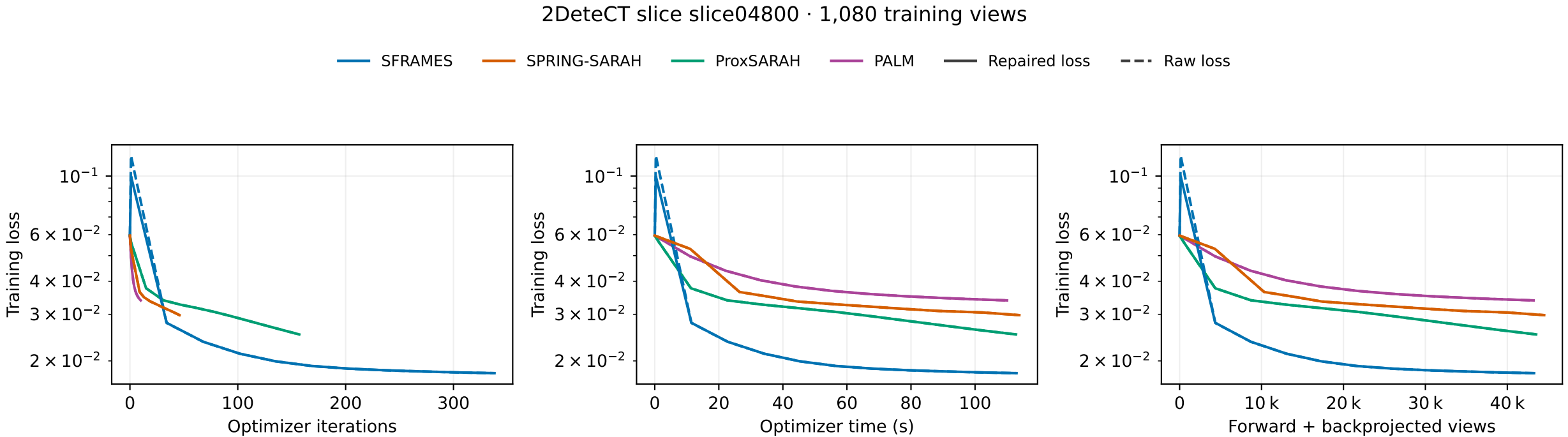
Held-out loss and reference error use repaired factors. Constraint violation and repair distance describe the raw iterate.



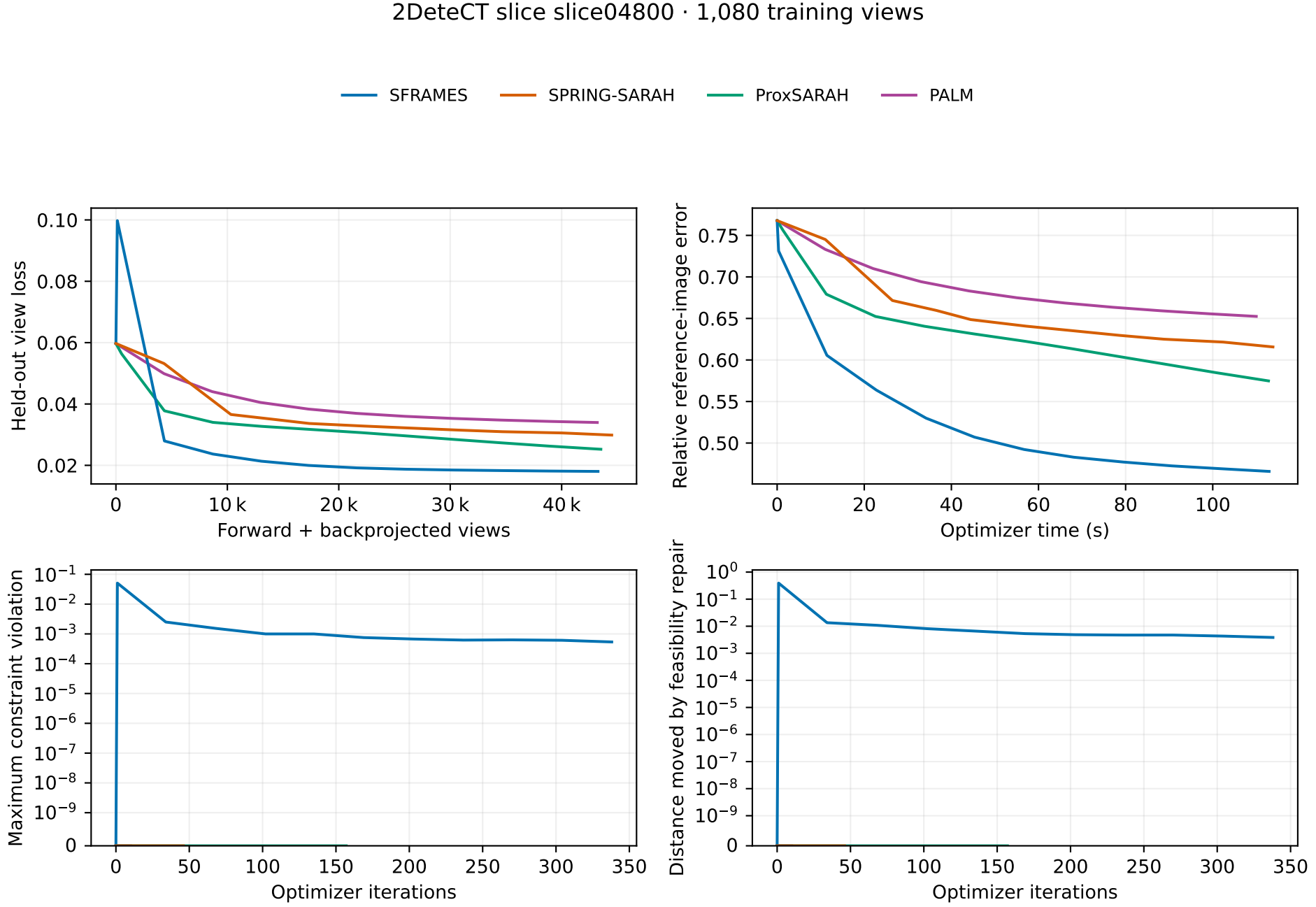
View counts include optimizer refreshes and recursive gradient differences. Timing excludes diagnostic probes and checkpoint I/O.



Held-out loss and reference error use repaired factors. Constraint violation and repair distance describe the raw iterate.



View counts include optimizer refreshes and recursive gradient differences. Timing excludes diagnostic probes and checkpoint I/O.



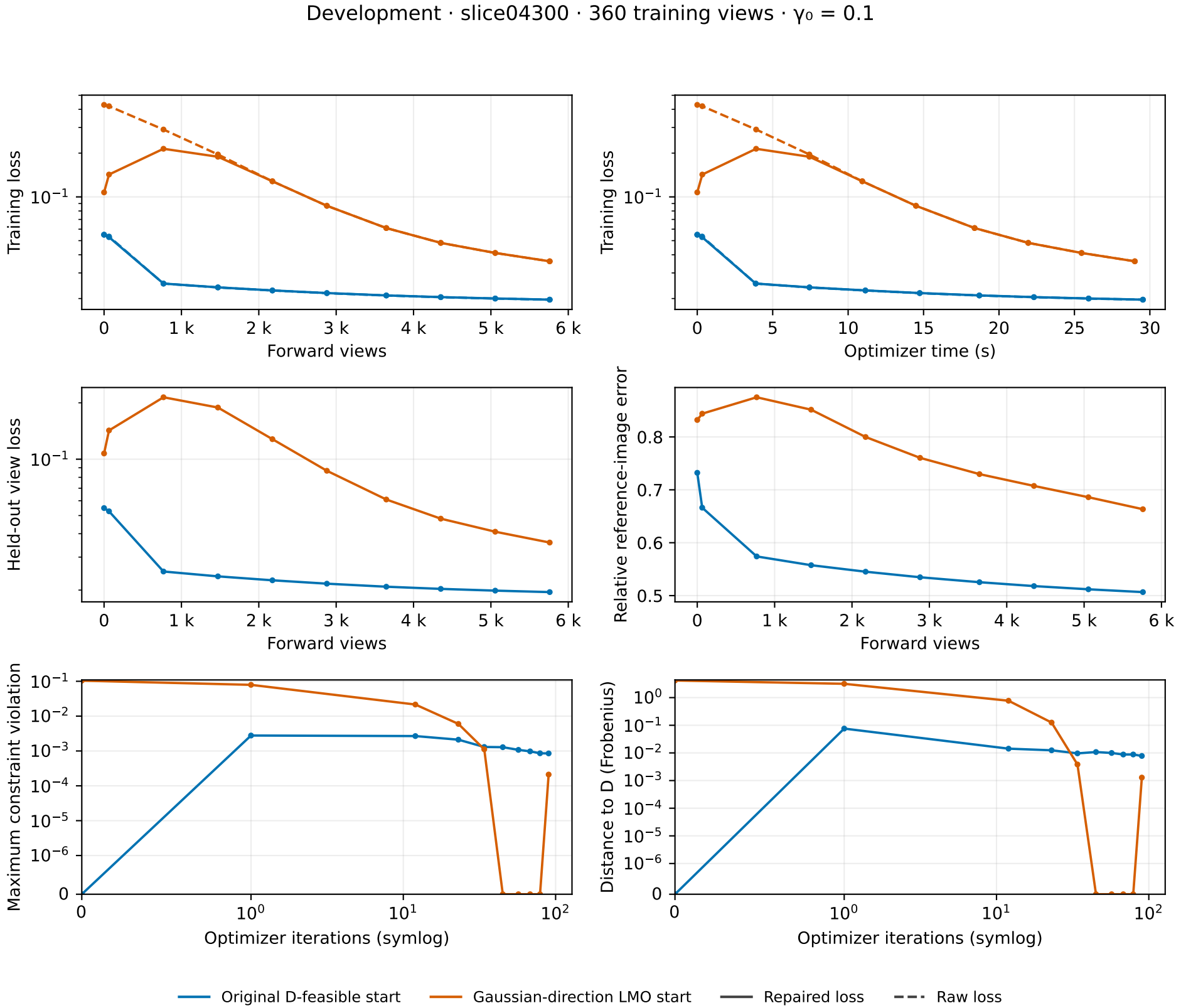
Held-out loss and reference error use repaired factors. Constraint violation and repair distance describe the raw iterate.

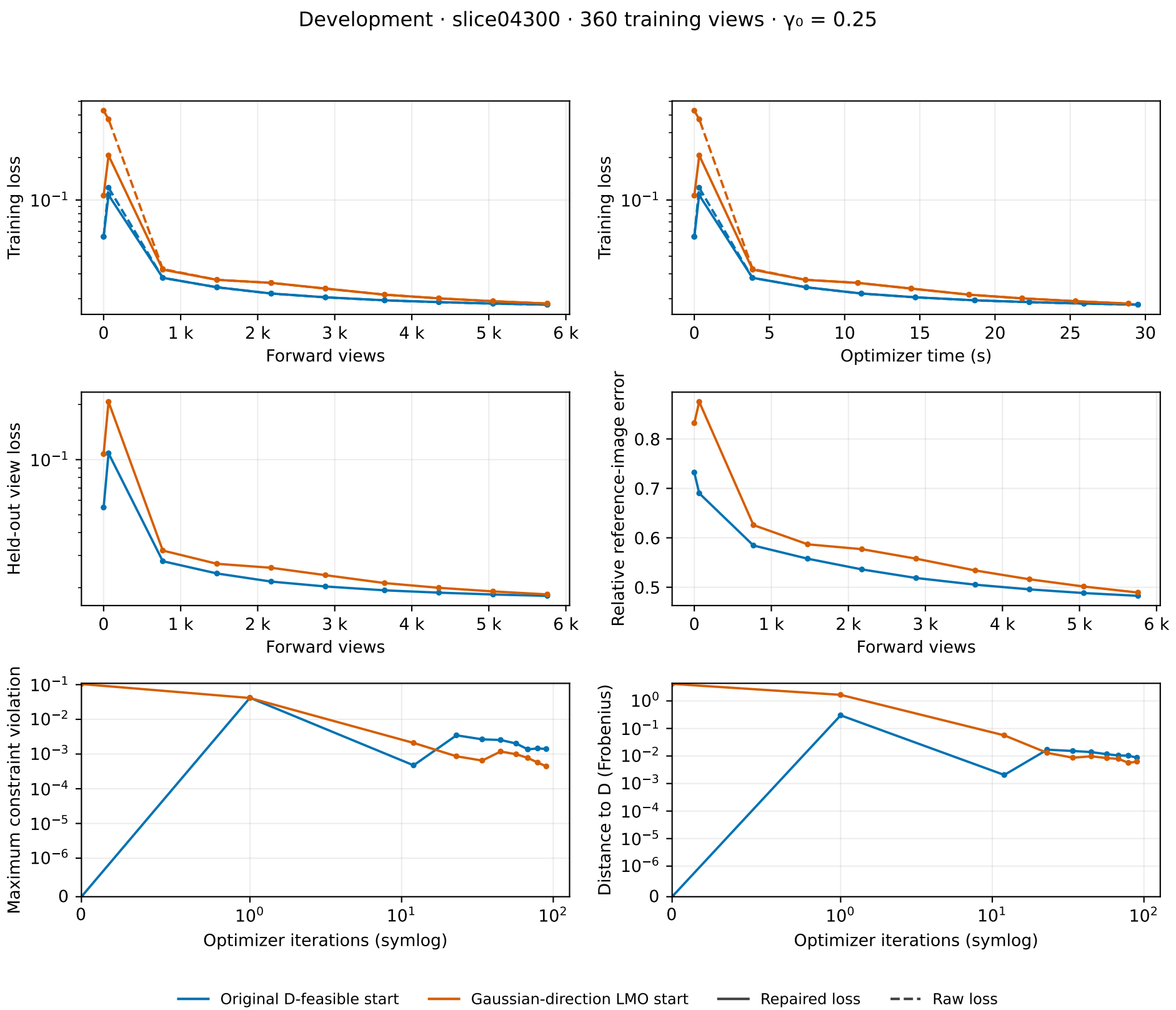
SFRAMES initialization ablation

The new start uses independent Gaussian directions and one Frobenius-ball LMO per block.

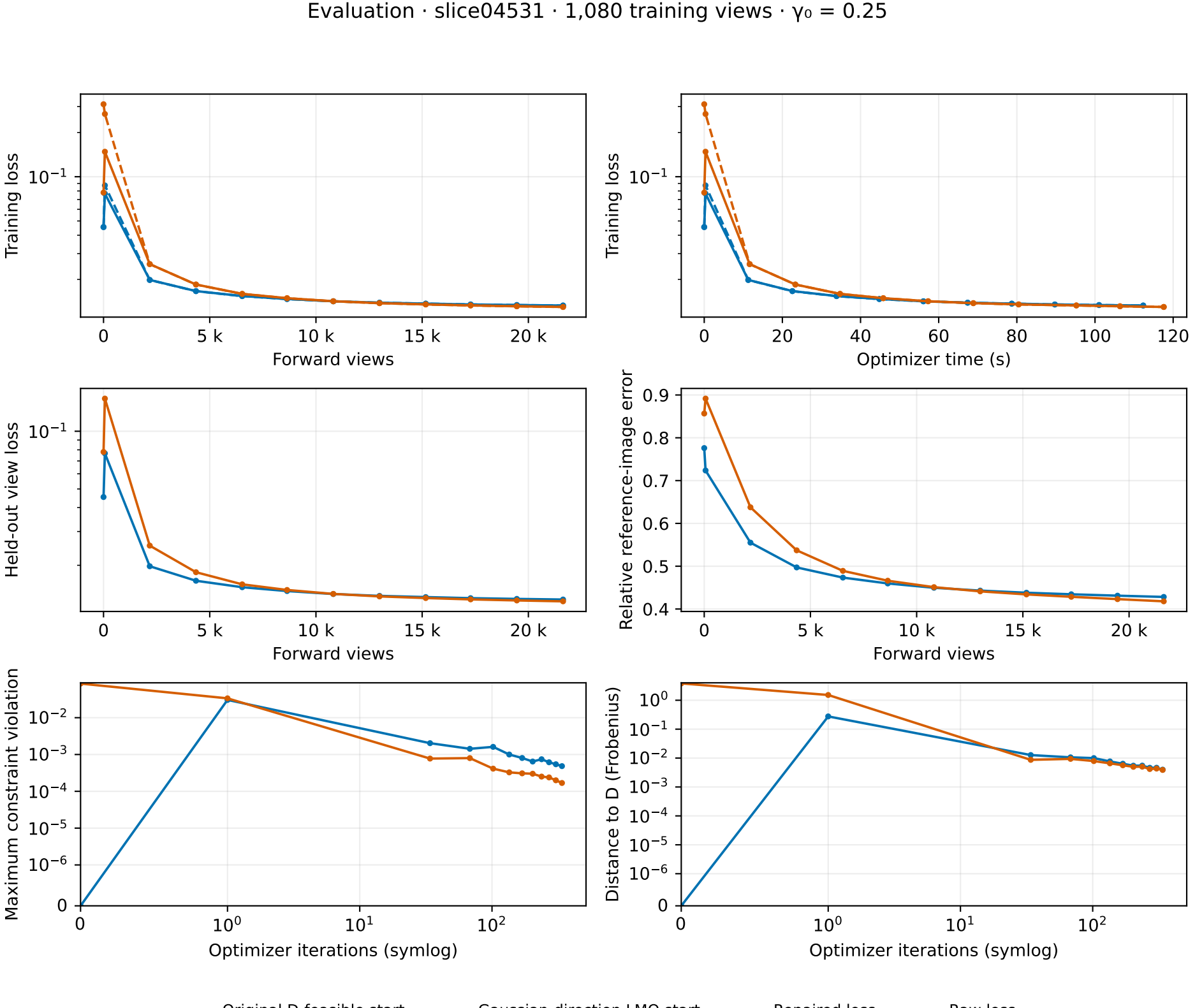
Slice / views	Phase · γ_0	New initial D distance	Original repaired loss	LMO repaired loss
slice04300 / 360	Development · 0.1	4.158	0.01965	0.03607
slice04300 / 360	Development · 0.25	4.158	0.01815	0.01847
slice04531 / 1080	Evaluation · 0.25	3.822	0.01329	0.01300
slice04531 / 360	Evaluation · 0.25	3.822	0.01328	0.01297
slice04800 / 1080	Evaluation · 0.25	4.235	0.01798	0.01771
slice04800 / 360	Evaluation · 0.25	4.235	0.01793	0.01767

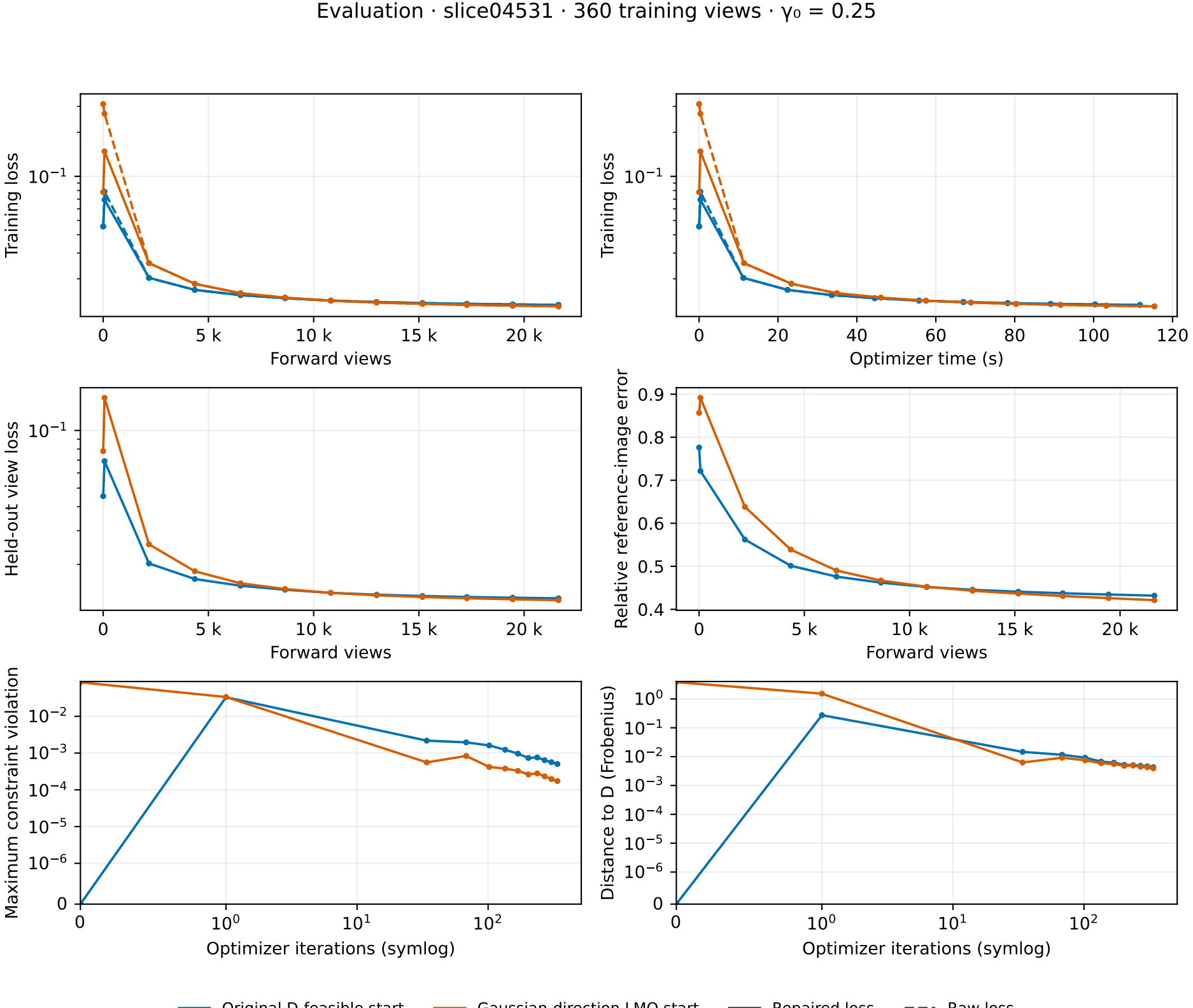
The loss columns use the latest common saved iteration with identical view counts.
The original initial D distance is zero in every pair.

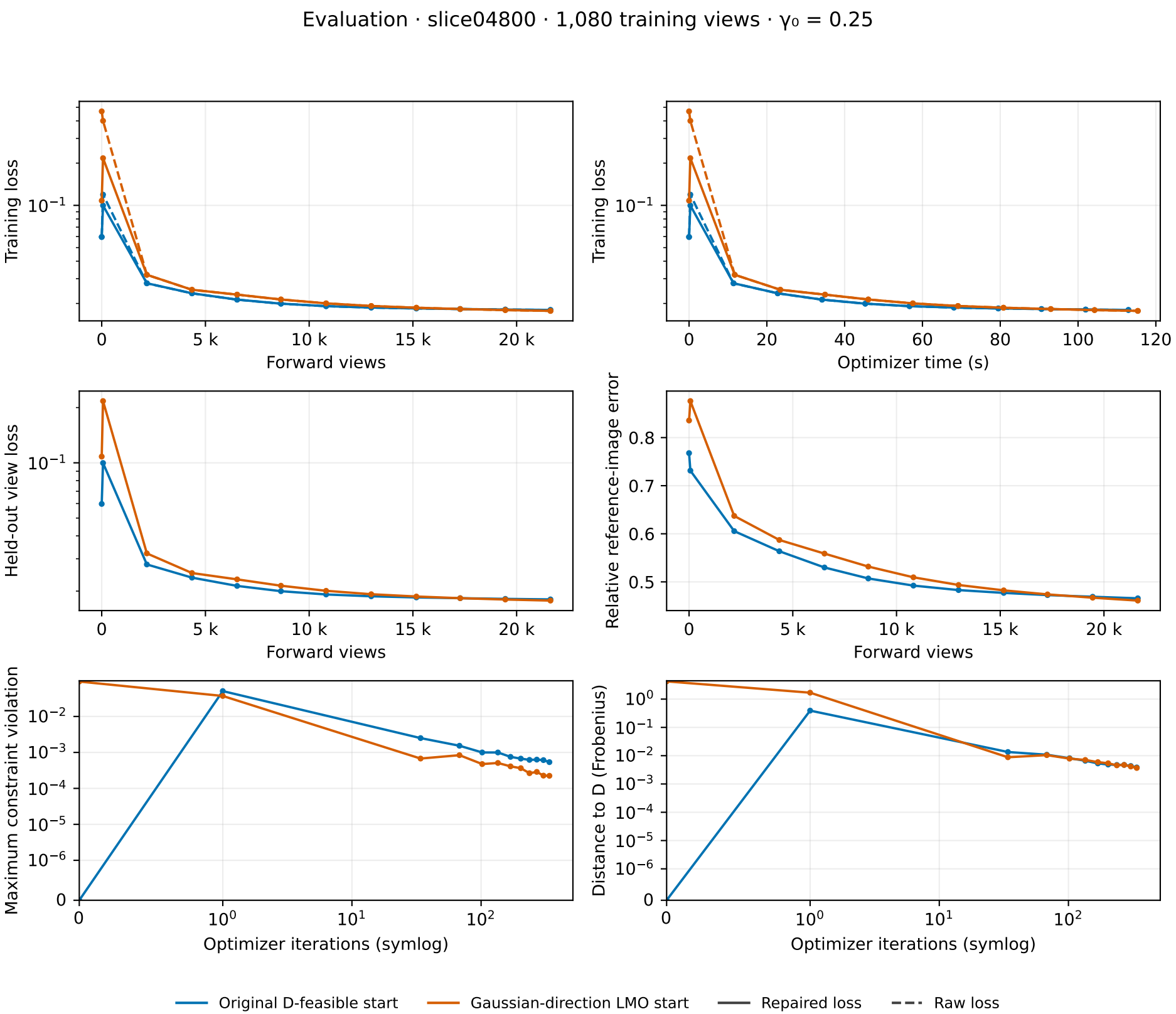

Development · slice04300 · 360 training views · γ₀ = 0.1
Training loss
Forward views
Optimizer time (s)
Held-out view loss
Relative reference-image error
Maximum constraint violation
Distance to D (Frobenius)
Optimizer iterations (symlog)
Original D-feasible start
Gaussian-direction LMO start
Repaired loss
Raw loss
Distance to D agrees with feasibility repair up to roundoff. The feasibility panels preserve zero and expand early iterations.

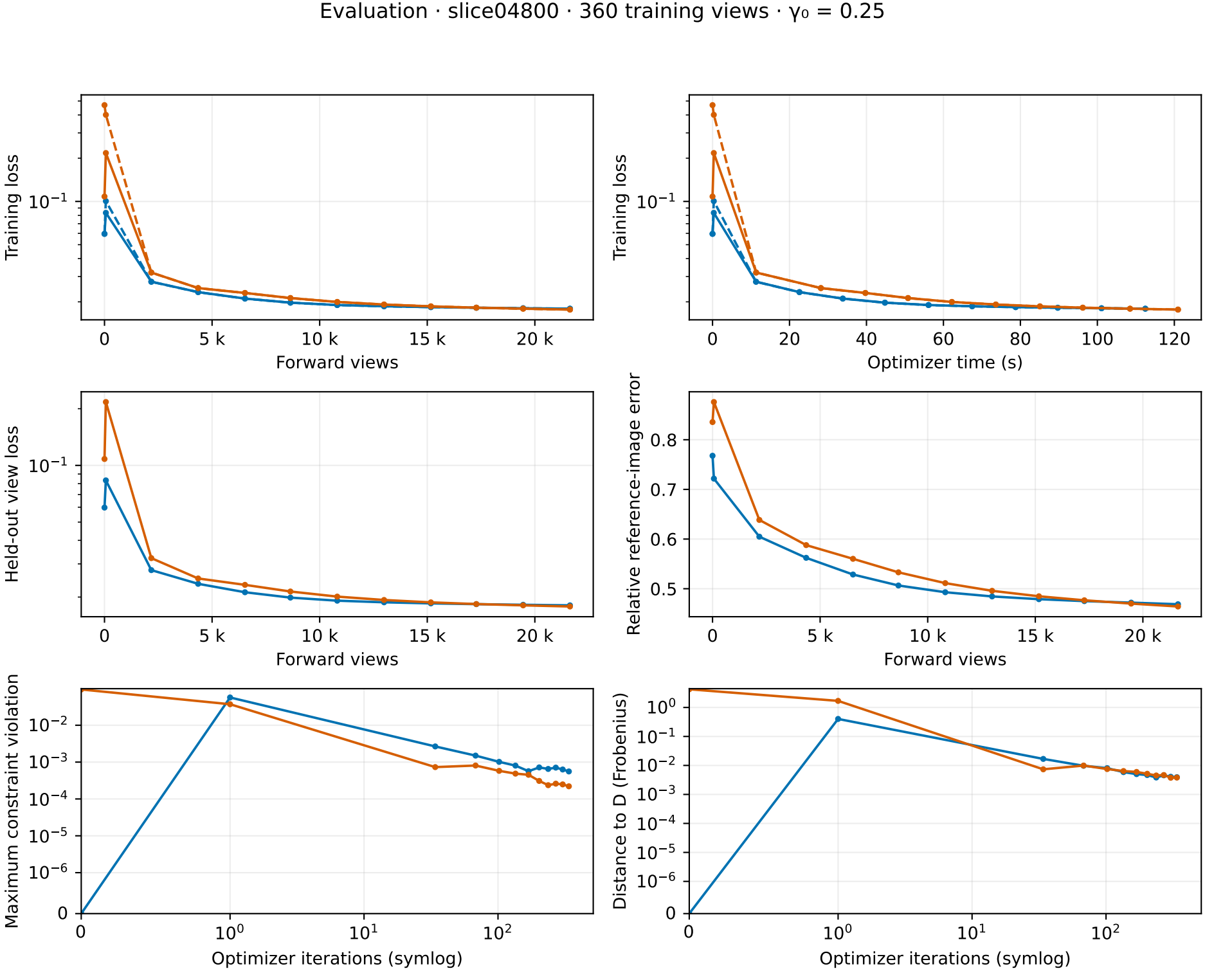


Distance to D agrees with feasibility repair up to roundoff. The feasibility panels preserve zero and expand early iterations.


Evaluation · slice04531 · 1,080 training views · γ₀ = 0.25
Training loss
Forward views
Optimizer time (s)
Held-out view loss
Relative reference-image error
Maximum constraint violation
Distance to D (Frobenius)
Optimizer iterations (symlog)
Original D-feasible start
Gaussian-direction LMO start
Repaired loss
Raw loss
Distance to D agrees with feasibility repair up to roundoff. The feasibility panels preserve zero and expand early iterations.


Evaluation · slice04531 · 360 training views · γ₀ = 0.25
Training loss
Forward views
Optimizer time (s)
Held-out view loss
Relative reference-image error
Maximum constraint violation
Distance to D (Frobenius)
Optimizer iterations (symlog)
Original D-feasible start
Gaussian-direction LMO start
Repaired loss
Raw loss
Distance to D agrees with feasibility repair up to roundoff. The feasibility panels preserve zero and expand early iterations.


Evaluation · slice04800 · 1,080 training views · γ₀ = 0.25
Training loss
Forward views
Optimizer time (s)
Held-out view loss
Relative reference-image error
Maximum constraint violation
Distance to D (Frobenius)
Optimizer iterations (symlog)
Original D-feasible start
Gaussian-direction LMO start
Repaired loss
Raw loss
Distance to D agrees with feasibility repair up to roundoff. The feasibility panels preserve zero and expand early iterations.


Evaluation · slice04800 · 360 training views · γ₀ = 0.25
Training loss
Forward views
Optimizer time (s)
Held-out view loss
Relative reference-image error
Maximum constraint violation
Distance to D (Frobenius)
Optimizer iterations (symlog)
Original D-feasible start
Gaussian-direction LMO start
Repaired loss
Raw loss
Distance to D agrees with feasibility repair up to roundoff. The feasibility panels preserve zero and expand early iterations.